

Information-Topological Theory: Minimal Foundational Formulation

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Abstract

We present the minimal foundational formulation of Information-Topological Theory (ITT) as a timeless canonical gauge framework on a three-dimensional configuration space. The formulation separates standard mathematical results from ITT-specific postulates, completed intermediate derivations, negative mathematical results, and future dynamical predictions.

The standard sector contains the $SU(2)$ gauge structure on S^3 , the canonical ITT constraint, its formal Chern–Simons state, the unit BPST topological sector, the associated Atiyah–Patodi–Singer spectral-flow construction, ordinary integer Chern–Simons level quantization, and the convention-fixed rational Chern–Simons invariant of the fundamental flat sector of $L(6,1) = S^3/\mathbb{Z}_6$. The level- k differential Chern–Simons evaluation of this sector gives the phase $\exp(2\pi i k \text{cs}(\rho_1)) = e^{-2\pi i k/6}$. A further analysis shows that the presently specified data do not canonically identify this phase with a stabilizer action on a hypothetical three-dimensional prequantum line fiber. ITT therefore formulates level selection as an explicit phase-triviality postulate rather than as an unproved equivariant-line-bundle theorem.

Under that theory-specific selection postulate, $Z_k(L(6,1), \rho_1) = 1$, one obtains $k \in 6\mathbb{Z}$ and hence $k_{\min} = 6$. No Standard-Model parameter fitting or generation-count argument is used in the foundational level selection.

1 Scope and Logical Status

The Information-Topological Theory (ITT) is formulated as a timeless canonical gauge theory on a three-dimensional spatial manifold. The fundamental configuration space is the gauge-orbit space

$$\mathcal{C}_{\text{ITT}} = \mathcal{A}(S^3)/\mathcal{G}(S^3), \quad (1)$$

where $\mathcal{A}(S^3)$ denotes the space of smooth $SU(2)$ connections on S^3 and $\mathcal{G}(S^3)$ the corresponding group of gauge transformations.

No preferred external physical time is postulated at the foundational level. A parameter used to label a family of configurations is therefore not identified with physical time.

The theory separates four logical levels:

- (i) **Standard mathematics:** gauge theory on S^3 , BPST topology, APS spectral flow, ordinary Chern–Simons level quantization, lens-space topology, and rational Chern–Simons invariants.
- (ii) **ITT defining structure:** the timeless canonical constraint and the phase-triviality requirement used for level selection.
- (iii) **Completed intermediate derivations:** the differential Chern–Simons phase and its induced cyclic character associated with the selected fundamental lens-space sector.
- (iv) **Open or future ITT results:** dynamical derivation of $N_{\text{gen}} = 3$, precision phenomenology, and any stronger physical interpretation of the selected level.

The explicit three-dimensional equivariant-line interpretation is not assumed as an established theorem. Paper 1 records the mathematical obstruction to obtaining such an identification from the presently specified data.

2 Canonical Configuration Space and Conventions

2.1 Gauge Variables

The canonical variables are

$$A_i^a(x), \quad E_a^i(x), \quad (2)$$

with fundamental Poisson brackets

$$\{A_i^a(x), E_b^j(y)\} = \delta_i^j \delta_b^a \delta^{(3)}(x - y), \quad (3)$$

and

$$\{A_i^a(x), A_j^b(y)\} = 0, \quad \{E_a^i(x), E_b^j(y)\} = 0. \quad (4)$$

The momentum E_a^i is taken to be a densitized spatial vector of weight +1. The $SU(2)$ generators are normalized by

$$T_a = \frac{\sigma_a}{2i}, \quad \text{tr}(T_a T_b) = -\frac{1}{2} \delta_{ab}, \quad [T_a, T_b] = \epsilon_{abc} T_c. \quad (5)$$

The curvature is

$$F = dA + A \wedge A, \quad (6)$$

and the magnetic density is

$$B_a^i = \frac{1}{2} \epsilon^{ijk} F_{jk,a}, \quad (7)$$

where ϵ^{ijk} denotes the spatial Levi-Civita density.

3 Chern–Simons Functional and Fundamental ITT Constraint

Define the unnormalized Chern–Simons functional

$$S_{\text{CS}}^{(0)}[A] = \int_{S^3} \text{tr} \left(A \wedge dA + \frac{2}{3} A \wedge A \wedge A \right). \quad (8)$$

With the trace convention above, its functional derivative is

$$\frac{\delta S_{\text{CS}}^{(0)}}{\delta A_i^a(x)} = -B_a^i(x). \quad (9)$$

The fundamental ITT constraint is

$$\boxed{\mathcal{C}_a^i \equiv E_a^i + \frac{k}{4\pi} B_a^i = E_a^i - \frac{k}{4\pi} \frac{\delta S_{\text{CS}}^{(0)}}{\delta A_i^a} \approx 0.} \quad (10)$$

The corresponding first-order action is

$$S_{\text{ITT}} = \int d\tau \int_{S^3} d^3x \left[E_a^i \frac{dA_i^a}{d\tau} - N_i^a \mathcal{C}_a^i - A_0^a \mathcal{G}_a \right]. \quad (11)$$

Here τ is an auxiliary parameter labeling a family of configurations; it is not assumed to be physical time.

4 Constraint Algebra

Proposition 4.1 (Commuting ITT constraints). *On the closed manifold S^3 , the constraints*

$$\mathcal{C}_a^i = E_a^i - \frac{k}{4\pi} \frac{\delta S_{\text{CS}}^{(0)}}{\delta A_i^a} \quad (12)$$

satisfy

$$\{\mathcal{C}_a^i(x), \mathcal{C}_b^j(y)\} = 0. \quad (13)$$

Proof. The term quadratic in functional derivatives vanishes because $\{A, A\} = 0$, while $\{E, E\} = 0$. The two mixed terms are opposite second functional derivatives of the same smooth functional $S_{\text{CS}}^{(0)}$. Since S^3 is closed, no boundary contribution remains, and equality of mixed functional derivatives gives exact cancellation. $\square$

The non-Abelian Bianchi identity gives

$$D_i B_a^i \equiv 0. \quad (14)$$

Therefore

$$D_i \mathcal{C}_a^i = D_i E_a^i \equiv \mathcal{G}_a. \quad (15)$$

Hence

$$\mathcal{G}_a = D_i \mathcal{C}_a^i. \quad (16)$$

Remark 4.2. *At the classical level the Gauss constraint is a differential consequence of the ITT constraint. The full set is naturally described as a first-class, differentially related constraint system. At the quantum level the corresponding identity requires a fixed gauge-covariant operator ordering.*

For the scalar constraint we use the densitized convention

$$\mathcal{H}_{\text{ITT}} = \frac{1}{2\sqrt{g}} g_{ij} \mathcal{C}_a^i \mathcal{C}^{ja}. \quad (17)$$

This definition fixes the density weight explicitly.

5 Formal Chern–Simons State

Canonical quantization is represented formally[3] by

$$\hat{E}_a^i = -i \frac{\delta}{\delta A_i^a}. \quad (18)$$

Define

$$\Psi_k[A] = \mathcal{N} \exp \left[i \frac{k}{4\pi} S_{\text{CS}}^{(0)}[A] \right]. \quad (19)$$

Then

$$\hat{E}_a^i \Psi_k[A] = -\frac{k}{4\pi} B_a^i \Psi_k[A], \quad (20)$$

so that formally

$$\hat{\mathcal{C}}_a^i \Psi_k[A] = 0. \quad (21)$$

Consequently, for the positive quadratic operator corresponding to the classical constraint,

$$\hat{\mathcal{H}}_{\text{ITT}} \Psi_k[A] = 0 \quad (22)$$

formally, subject to a specified regularization and operator ordering.

This is a formal exact solution of the ITT constraint system. A complete Hilbert-space construction, functional measure, self-adjoint realization, and ultraviolet regularization are separate analytical problems and are not claimed to be solved here.

6 Unit Topological Sector: BPST and APS

Consider the standard $SU(2)$ BPST instanton of scale ρ on flat $\mathbb{R}^4$, restricted to the four-ball D_R^4 [1]. Define

$$c = \frac{R^2}{R^2 + \rho^2}. \quad (23)$$

With the normalization above,

$$Q(R) = -\frac{1}{8\pi^2} \int_{D_R^4} \text{tr}(F \wedge F) = 3c^2 - 2c^3 = \frac{R^4(R^2 + 3\rho^2)}{(R^2 + \rho^2)^3}. \quad (24)$$

On the boundary,

$$A|_{S_R^3} = c g^{-1} dg, \quad (25)$$

and the Chern–Simons transgression gives, with the same orientation convention,

$$\frac{1}{2\pi} S_{\text{CS}}(A|_{S_R^3}) = 3c^2 - 2c^3. \quad (26)$$

Thus

$$\lim_{R/\rho \rightarrow \infty} Q(R) = 1. \quad (27)$$

Introduce the auxiliary cylinder

$$Y = [0, 1] \times S^3 \quad (28)$$

with product metric and the family

$$A(c) = c g^{-1} dg. \quad (29)$$

The parameter c labels a path in configuration space and is not physical time. For the unit-winding interpolation, the standard APS spectral-flow correspondence identifies the net spectral flow with the unit topological index[2]:

$$\boxed{\text{SF}(\mathcal{D}_{A(c)}) = \text{ind}(\mathcal{D}_Y^{\text{APS}}) = 1.} \quad (30)$$

The statement above concerns the net spectral flow. It does not assert, without a separate spectral calculation, that exactly one individual eigenvalue crosses zero.

7 Standard Chern–Simons Level Quantization

For a large gauge transformation

$$w \in \pi_3(SU(2)) \cong \mathbb{Z}, \quad (31)$$

the normalized Chern–Simons amplitude is gauge invariant only when the level satisfies

$$\boxed{k \in \mathbb{Z}.} \quad (32)$$

This is the standard Chern–Simons quantization condition[3]. It does not select $k = 6$.

8 Lens-Space Sector

The ITT quotient geometry is defined by the free diagonal action

$$(z_1, z_2) \longmapsto (e^{2\pi i/6} z_1, e^{2\pi i/6} z_2) \quad (33)$$

on $S^3 \subset \mathbb{C}^2$. Thus

$$\boxed{L(6, 1) = S^3/\mathbb{Z}_6,} \quad (34)$$

and

$$\pi_1(L(6, 1)) \cong \mathbb{Z}_6. \quad (35)$$

Let ρ_1 be the fundamental non-trivial flat representation

$$\rho_1(\gamma) = \begin{pmatrix} e^{2\pi i/6} & 0 \\ 0 & e^{-2\pi i/6} \end{pmatrix}, \quad (36)$$

for a chosen generator γ of $\pi_1(L(6, 1))$.

Define the normalized Chern–Simons invariant by

$$\text{cs}(A) = \frac{1}{8\pi^2} \int_{L(6,1)} \text{tr} \left(A \wedge dA + \frac{2}{3} A \wedge A \wedge A \right) \pmod{\mathbb{Z}}. \quad (37)$$

With the orientation and trace convention fixed above, the fundamental flat sector has

$$\boxed{\text{cs}(\rho_1) = -\frac{1}{6} \pmod{\mathbb{Z}}.} \quad (38)$$

One way to obtain this value is through the (-6) -framed surgery description of $L(6, 1)$ [4, 5]. The inverse surgery matrix gives the rational boundary linking pairing $-1/6$. The fractional value is a three-dimensional secondary invariant.

Remark 8.1. *The rational value $-1/6$ is not an ordinary second Chern number on a closed four-manifold such as $L(6, 1) \times S^1$. Ordinary integral characteristic numbers remain integral.*

9 Differential Chern–Simons Phase

The level- k characteristic class is the integral class

$$kc_2 \in H^4(BSU(2); \mathbb{Z}), \quad (39)$$

with a differential refinement in the sense of Cheeger–Simons differential characters [7, 6]. For the flat connection (P_{ρ_1}, A_{ρ_1}) on the closed three-manifold $L(6, 1)$, its evaluation gives the exponentiated Chern–Simons invariant

$$\boxed{Z_k(L(6, 1), \rho_1) = \exp(2\pi i k \text{cs}(\rho_1)).} \quad (40)$$

Therefore

$$\boxed{Z_k(L(6, 1), \rho_1) = e^{-2\pi i k/6}.} \quad (41)$$

This is a derived consequence of the standard differential Chern–Simons construction together with the convention-fixed lens-space invariant; it is not an ITT postulate.

10 Cyclic Character Induced by the Fundamental Phase

Let

$$\Gamma = \mathbb{Z}_6 = \langle \zeta \mid \zeta^6 = 1 \rangle. \quad (42)$$

Since $k \in \mathbb{Z}$,

$$Z_k^6 = e^{-2\pi i k} = 1. \quad (43)$$

Hence the phase canonically determines, relative to the chosen generator ζ , a cyclic character

$$\boxed{\chi_k^{\text{CS}} : \Gamma \longrightarrow U(1), \quad \chi_k^{\text{CS}}(\zeta^n) = Z_k^n.} \quad (44)$$

In particular,

$$\boxed{\chi_k^{\text{CS}}(\zeta) = e^{-2\pi i k/6}.} \quad (45)$$

No statement is made that this character equals the stabilizer action of a separately constructed three-dimensional prequantum line; that identification is addressed as a negative result in the level-selection paper.

11 Revised ITT Level-Selection Postulate

The standard Chern–Simons construction supplies the phase Z_k , but does not by itself supply a canonical identification of that phase with a fiber character of a three-dimensional line object. ITT therefore makes the following explicit theory-specific choice.

Postulate 11.1 (ITT Fundamental-Sector Phase Triviality). *The fundamental quotient sector $L(6,1)$ is admitted by ITT at level k only when its level- k Chern–Simons phase is trivial:*

$$\boxed{Z_k(L(6,1), \rho_1) = 1.} \quad (46)$$

This postulate is stronger and more transparent than assuming an unconstructed equivariant line-bundle lift. It is an ITT-specific selection rule and is not claimed to follow from ordinary $SU(2)$ Chern–Simons theory.

12 Conditional Level Selection

Using the derived phase,

$$Z_k = e^{-2\pi i k/6}, \quad (47)$$

the phase-triviality postulate gives

$$e^{-2\pi i k/6} = 1. \quad (48)$$

Therefore

$$\boxed{k \in 6\mathbb{Z}.} \quad (49)$$

For positive integer level,

$$\boxed{k_{\min} = 6.} \quad (50)$$

The logical chain is therefore

$$\text{standard CS quantization } (k \in \mathbb{Z}) + \text{ITT phase-triviality postulate} \implies k_{\min} = 6. \quad (51)$$

13 Independent Soliton Sector

The number of generations is not used in the level-selection argument. ITT treats

$$N_{\text{gen}} = 3 \quad (52)$$

as an independent dynamical hypothesis to be tested by a future soliton analysis. The objective is to construct an ITT energy functional $E_{\text{ITT}}[n, A]$ for which the stable topological sectors satisfy

$$Q_H = 1, 2, 3, \quad (53)$$

while higher-charge configurations are unstable or energetically decomposable. Until such a stability theorem is obtained, $N_{\text{gen}} = 3$ remains an open ITT result and is not used to derive $k = 6$.

14 Foundational Logical Chain

The minimal ITT foundation is summarized by

$$\begin{aligned}
& \mathcal{C}_{\text{ITT}} = \mathcal{A}(S^3)/\mathcal{G}(S^3) \\
& \Downarrow \\
& \boxed{\mathcal{C}_a^i = E_a^i + \frac{k}{4\pi} B_a^i, \quad \{\mathcal{C}, \mathcal{C}\} = 0} \\
& \Downarrow \\
& \boxed{D_i \mathcal{C}_a^i = \mathcal{G}_a} \\
& \Downarrow \\
& \boxed{\Psi_k[A] = \mathcal{N} e^{ik S_{\text{CS}}^{(0)}[A]/4\pi}} \\
& \Downarrow \\
& \boxed{Q_{\text{BPST}} \rightarrow 1, \quad \text{SF} = \text{ind}_{\text{APS}} = 1} \\
& \Downarrow \\
& \boxed{L(6, 1) = S^3/\mathbb{Z}_6, \quad \text{cs}(\rho_1) = -\frac{1}{6}} \\
& \Downarrow \\
& \boxed{Z_k = e^{2\pi i k \text{cs}(\rho_1)} = e^{-2\pi i k/6}} \\
& \Downarrow \\
& \boxed{Z_k = 1 \quad (\text{ITT phase-triviality postulate})} \\
& \Downarrow \\
& \boxed{k \in 6\mathbb{Z}} \\
& \Downarrow \\
& \boxed{k_{\min} = 6.}
\end{aligned} \tag{54}$$

The independent soliton problem begins after this foundational chain:

$$E_{\text{ITT}} \longrightarrow \text{stable Hopf sectors} \longrightarrow N_{\text{gen}} = 3. \tag{55}$$

15 Foundational Status and Open Problems

The present version deliberately does not claim completion of the full physical theory. The current status is:

Object	Status	Role
$SU(2)$ gauge geometry on S^3	Established	Foundation
BPST unit sector	Established	Topological input
APS spectral-flow framework	Established	Unit-index construction
$k \in \mathbb{Z}$	Established	Standard CS quantization
$L(6, 1) = S^3/\mathbb{Z}_6$	Established	ITT quotient geometry
$\text{cs}(\rho_1) = -1/6$	Derived/Convention-fixed	Level-selection input
$Z_k = e^{-2\pi i k/6}$	Derived	Differential-CS evaluation
χ_k^{CS}	Derived construction	Cyclic character
3D equivariant fiber identification	Obstructed	Paper 1 negative result
$Z_k = 1$	ITT Postulate	Level selection
$k_{\min} = 6$	Conditional	Consequence of postulate
$N_{\text{gen}} = 3$	Open	Soliton paper target
α , masses, $g - 2$, cosmology	Future	Prediction modules

The foundational theory therefore makes the restricted claim that standard differential Chern–Simons mathematics supplies the phase of the selected lens-space sector, while the phase-triviality condition is an explicit ITT selection choice. No claim is made that $k = 6$ is a universal theorem of ordinary Chern–Simons theory.

References

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